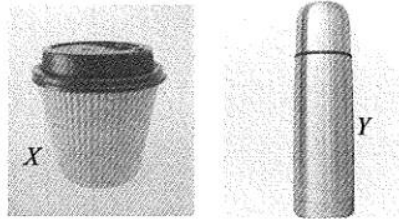


Book 1 section 1.4: DSE multiple choice

2014 Q1

1.



Two identical scoops of ice-cream are transferred from a refrigerator into paper cup X and vacuum flask Y shown above. Under room temperature, the time required for the ice-cream in the containers to melt completely is t_X and t_Y respectively. What is the expected result and explanation?

- A. $t_X > t_Y$ as the vacuum flask reduces heat loss to the surroundings.
- B. $t_X > t_Y$ as the vacuum flask retains the heat.
- C. $t_Y > t_X$ as the vacuum flask keeps things cold by releasing heat into the surroundings.
- D. $t_Y > t_X$ as the vacuum flask reduces the rate of heat gain from the surroundings.

2016 Q1

1. Some icy cold liquid is kept cold inside a vacuum flask. Which statements are correct?
- (1) The flask's cork stopper reduces heat gain from the surroundings.
 - (2) The silver coating on the inner surface of the glass wall is a good reflector of infra-red.
 - (3) The vacuum between the double glass walls reduces heat gain by radiation.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2015 Q1

1. A driver turns off the engine after parking his car outdoors under the sun. Two hours later when getting into the car, he feels that the inside of the car is far hotter than outside. The best explanation is
- A. the car's engine is still generating heat after the engine has been switched off.
 - B. the car's metal parts absorb infra-red radiation at a faster rate than the surroundings.
 - C. the glass windows of the car trap infra-red radiation and a greenhouse effect results.
 - D. the surrounding air is a good insulator of heat which reduces heat loss by conduction.